

Due on Tuesday, February 28th.

If pictures are needed/relevant, please provide them with your solutions.

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This problem set is about the relationships between electric field, electric potential and electric charge density.

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1. **[4 pts.]** Using cartesian coordinates, show that the curl of the gradient of a scalar field is zero, i.e., $\nabla \times (\nabla f) = 0$ for any scalar field $f(x, y, z)$.

[Note: this means: a vector field that can be expressed as a gradient of a scalar field, must have zero curl. Thus the electric field, which is the gradient of $-V$, must have zero curl: $\nabla \times \mathbf{E} = 0$.]

2. The electric potential in some region of space is given by

$$V(x, y, z) = \frac{C}{\sqrt{x^2 + y^2 + z^2}}$$

or, in spherical coordinates, $V(r) = C/r$. Here C is a positive constant.

- (a) **[5 pts.]** Calculate an expression for the electric field in this region, using cartesian coordinates.
 - (b) **[4 pts.]** Noting that $x\hat{i} + y\hat{j} + z\hat{k} = \mathbf{r} = r\hat{r}$, express the expression for $\mathbf{E}$ in spherical coordinates. Here $\hat{r}$ is the radial unit vector of spherical coordinates, sometimes also denoted as $\mathbf{e}_r$ or $\hat{e}_r$.
 - (c) **[4 pts.]** Look up the expression for the gradient operator in spherical coordinates. Re-derive the expression for $\mathbf{E}$, now directly in spherical coordinates.
 - (d) **[5 pts.]** Look up the expression for the Laplacian operator in spherical coordinates. By calculating the Laplacian of V , find out the charge density ρ in this region.
 - (e) **[5 pts.]** Calculate the Laplacian of V (and hence the charge density ρ) using cartesian coordinates. (This is a cumbersome calculation.)
3. **[5 pts.]** Show that the vector field $\mathbf{A} = y\hat{i} - x\hat{j}$ cannot be a valid electric field, because it's curl is nonzero. Which direction does $\nabla \times \mathbf{A}$ point in?

4. The electric field is claimed to be $\mathbf{E} = x\hat{i} + y\hat{j}$ in some region of space.
- (a) **[5 pts.]** Show that this satisfies the requirement of zero curl.
 - (b) **[5 pts.]** Using Maxwell's first equation, find the electric charge density ρ in this region of space. How does ρ vary in space?
 - (c) **[3 pts.]** Calculate the difference of electric potential between the points $(1, 5, -1)$ and $(1, 5, +1)$.
 - (d) **[5 pts.]** Calculate the difference of electric potential between the points $(0, 5, 0)$ and $(2, 5, 0)$.